

MCGRATH, AK, USA

WMO: 702310

Lat: 62.957N

Lon: 155.610W

Elev: 102

StdP: 100.11

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26510

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.0	-39.9	-37.7	0.1	-33.8	-36.3	0.1	-32.9	8.3	-8.1	7.4	-11.2	0.2	280	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	26.0	15.9	24.0	15.0	22.1	14.3	17.1	23.5	16.1	21.9	15.2	20.3	2.7	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	10.7	18.3	13.8	10.0	17.4	12.9	9.4	16.7	48.3	23.4	45.3	21.9	42.7	20.4	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.2	5.3	DB	-44.4	28.7	3.2	1.9	-46.7	30.1	-48.6	31.2	-50.4	32.2	-52.7	33.6	(4)
(5)			WB	-39.1	19.2	3.8	1.4	-41.8	20.3	-44.1	21.1	-46.2	21.9	-48.9	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.9	-21.1	-15.0	-11.4	-0.1	8.8	14.4	15.6	12.9	7.7	-2.1	-13.6	-19.4	(6)	
	DBStd	14.83	11.36	9.20	7.56	5.75	4.20	2.87	2.74	3.04	3.66	6.07	8.40	9.59	(7)	
	HDD10.0	4792	964	699	664	304	70	2	1	8	84	376	709	910	(8)	
	HDD18.3	7389	1222	933	922	554	295	121	92	170	319	634	959	1168	(9)	
	CDD10.0	456	0	0	0	0	33	136	174	98	15	1	0	0	(10)	
	CDD18.3	13	0	0	0	0	0	4	7	2	0	0	0	0	(11)	
	CDH23.3	232	0	0	0	0	14	87	105	26	0	0	0	0	(12)	
	CDH26.7	32	0	0	0	0	1	15	14	3	0	0	0	0	(13)	
(14) Wind	WSAvg	1.8	1.1	1.4	2.0	2.3	2.5	2.2	2.0	1.9	1.9	1.8	1.3	1.0	(14)	
(15) Precipitation	PrecAvg	428	24	18	18	20	23	40	59	66	58	37	35	33	(15)	
	PrecMax	654	95	94	72	71	78	88	136	137	157	118	110	128	(16)	
	PrecMin	177	2	0	0	1	2	4	10	5	5	4	1	3	(17)	
	PrecStd	105	20	18	18	18	14	20	31	25	40	22	26	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.4	5.0	7.5	15.5	25.3	28.5	28.4	26.5	20.3	12.6	5.2	3.6	(19)	
		MCWB	1.4	1.9	2.6	7.1	12.9	15.9	17.2	17.6	12.5	8.3	2.8	1.1	(20)	
	2%	DB	1.2	2.6	4.5	12.9	22.0	25.7	26.2	23.0	17.6	9.2	2.3	0.6	(21)	
		MCWB	0.0	1.0	1.2	5.9	11.8	15.3	16.4	15.7	11.2	6.3	1.2	-0.5	(22)	
	5%	DB	-1.1	0.4	2.5	10.7	19.5	23.6	24.1	20.6	15.7	7.3	0.4	-2.6	(23)	
		MCWB	-2.0	-0.9	0.0	4.7	10.5	14.2	15.8	14.5	10.4	5.1	-0.3	-3.5	(24)	
	10%	DB	-5.5	-2.4	0.3	8.7	17.0	21.7	22.1	18.5	13.8	5.5	-2.1	-5.9	(25)	
		MCWB	-6.0	-3.5	-1.8	3.7	9.3	13.4	15.2	13.5	9.9	3.8	-2.8	-6.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	2.9	3.3	7.9	13.8	18.2	18.6	18.7	13.3	8.9	3.3	1.5	(27)	
		MCDB	3.7	4.3	5.9	13.9	22.9	25.7	25.9	24.5	17.7	12.0	4.4	2.9	(28)	
	2%	WB	0.4	1.0	1.7	6.3	12.5	16.2	17.4	16.7	12.1	6.8	1.2	-0.3	(29)	
		MCDB	1.5	2.6	4.1	11.8	20.5	23.6	24.0	21.6	16.0	8.7	2.3	1.1	(30)	
	5%	WB	-1.5	-0.8	0.2	5.1	11.2	15.1	16.5	15.4	11.2	5.4	-0.2	-3.2	(31)	
		MCDB	-0.3	0.6	2.4	10.2	17.8	21.9	22.5	19.3	14.5	7.2	0.5	-2.1	(32)	
	10%	WB	-4.8	-3.4	-1.8	4.0	10.1	14.1	15.6	14.3	10.3	3.9	-2.7	-6.1	(33)	
		MCDB	-4.1	-2.2	0.3	8.4	16.1	20.2	20.9	17.5	13.3	5.4	-2.0	-5.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	11.0	13.7	11.6	11.4	10.9	9.5	8.9	8.8	6.9	7.6	8.4	(35)	
		MCDBR	9.5	9.9	12.7	14.1	16.0	15.1	14.4	13.0	12.9	8.2	6.2	8.8	(36)	
	5% WB	MCWBR	8.2	8.3	9.8	8.1	7.1	6.5	6.3	6.6	7.3	5.6	5.5	7.8	(37)	
		MCWBR	9.3	9.7	11.1	12.7	14.2	13.2	12.2	10.8	10.6	7.3	6.1	8.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.200	0.260	0.269	0.319	0.357	0.373	0.372	0.372	0.312	0.284	0.213	0.188	(40)	
		taud	2.028	2.217	2.272	2.229	2.378	2.355	2.405	2.415	2.563	2.445	2.027	1.914	(41)	
	Edn at Noon	Edn at Noon	460	696	845	862	858	847	838	803	804	676	414	233	(42)	
		Edh at Noon	31	60	83	109	104	109	101	92	65	49	30	17	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.25	0.97	2.79	4.61	5.17	5.31	4.55	3.47	2.29	0.98	0.33	0.11	(44)	
	RadStd	RadStd	0.03	0.10	0.26	0.35	0.51	0.36	0.40	0.35	0.32	0.12	0.05	0.01	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.00	N/A	+0.85	N/A	N/A	N/A	-299	-358	N/A	N/A	(46)
(47)	Regional (5 neighbors)	+1.00	N/A	+0.85	N/A	N/A	N/A	-299	-358	N/A	N/A	(47)

Nomenclature: See separate page